

# Supplementary Materials

Please note that additional detailed Supplementary Materials (including full code, case studies, and results data) have been hosted in this separate, dedicated Supplementary Materials repository to facilitate access:

<https://anonymous.4open.science/r/CRAVE-D983>

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## Section S1: Detailed Statistical Tables

This section provides the comprehensive statistical results of the  $2 \times 2$  factorial evaluation.

### S1.1 LLM-as-a-Judge Results

Tables S1 to S3 detail the main effects across the **AllMean Pool**, while Tables S4 to S6 detail the main effects within the **Top-3 Pool**. For extended statistical results, including interaction effects, simple effects, and raw score distributions, please refer to the provided repository.

Table S1: Detailed Main Effects of the  $2 \times 2$  Factorial Evaluation (**AllMean Pool**, **Mistral-7B** Judge). Statistical significance is derived from the Mann-Whitney U test. Significant  $p$ -values ( $p < 0.05$ ) are highlighted in bold.

| Evaluation Metric | Analyzed Effect                 | Diff.<br>[95% CI]          | Cohen's $d$ | Statistic ( $U$ ) | $p$ -value   |
|-------------------|---------------------------------|----------------------------|-------------|-------------------|--------------|
| Relevance (RE)    | Factor A (Guidance: RAF vs AVL) | 0.004<br>[-0.081, 0.088]   | 0.008       | 32385.5           | 0.951        |
|                   | Factor B (Critic: w/o vs w/)    | -0.079<br>[-0.163, 0.005]  | -0.164      | 29623.5           | 0.068        |
| Coherence (CH)    | Factor A (Guidance: RAF vs AVL) | -0.194<br>[-0.409, 0.022]  | -0.160      | 32314.5           | 0.981        |
|                   | Factor B (Critic: w/o vs w/)    | 0.008<br>[-0.204, 0.221]   | 0.007       | 31641.5           | 0.421        |
| Emotion (EM)      | Factor A (Guidance: RAF vs AVL) | -0.056<br>[-0.091, -0.022] | -0.280      | 29561.5           | <b>0.001</b> |
|                   | Factor B (Critic: w/o vs w/)    | 0.034<br>[-0.001, 0.069]   | 0.168       | 34097.5           | <b>0.039</b> |
| Suspense (SU)     | Factor A (Guidance: RAF vs AVL) | -0.020<br>[-0.100, 0.061]  | -0.042      | 31403.0           | 0.524        |
|                   | Factor B (Critic: w/o vs w/)    | -0.012<br>[-0.092, 0.069]  | -0.026      | 32165.0           | 0.906        |
| Engagement (EG)   | Factor A (Guidance: RAF vs AVL) | -0.060<br>[-0.128, 0.009]  | -0.152      | 29796.0           | <b>0.043</b> |
|                   | Factor B (Critic: w/o vs w/)    | -0.026<br>[-0.095, 0.042]  | -0.066      | 31705.0           | 0.613        |
| Complexity (CX)   | Factor A (Guidance: RAF vs AVL) | -0.010<br>[-0.042, 0.022]  | -0.055      | 31783.0           | 0.443        |
|                   | Factor B (Critic: w/o vs w/)    | 0.010<br>[-0.021, 0.042]   | 0.057       | 32866.5           | 0.421        |
| Overall (OV)      | Factor A (Guidance: RAF vs AVL) | -0.027<br>[-0.064, 0.011]  | -0.122      | 30280.0           | 0.200        |
|                   | Factor B (Critic: w/o vs w/)    | -0.015<br>[-0.053, 0.023]  | -0.068      | 30881.0           | 0.357        |

*Note:* Mean Difference for Factor A =  $\text{Mean}_{\text{RAF}} - \text{Mean}_{\text{AVL}}$ . Mean Difference for Factor B =  $\text{Mean}_{\text{w/o\_Critic}} - \text{Mean}_{\text{w/_Critic}}$ . [95% CI] represents the 95% confidence interval of the mean difference.

Table S2: Detailed Main Effects of the  $2 \times 2$  Factorial Evaluation (**AllMean Pool**, **Gemma3-12B** Judge). Statistical significance is derived from the Mann-Whitney U test. Significant  $p$ -values ( $p < 0.05$ ) are highlighted in bold.

| Evaluation Metric | Analyzed Effect                 | Diff.<br>[95% CI]          | Cohen's $d$ | Statistic ( $U$ ) | $p$ -value   |
|-------------------|---------------------------------|----------------------------|-------------|-------------------|--------------|
| Relevance (RE)    | Factor A (Guidance: RAF vs AVL) | -0.007<br>[-0.051, 0.037]  | -0.028      | 32064.5           | 0.749        |
|                   | Factor B (Critic: w/o vs w/)    | -0.002<br>[-0.046, 0.041]  | -0.009      | 32254.0           | 0.916        |
| Coherence (CH)    | Factor A (Guidance: RAF vs AVL) | 0.088<br>[0.037, 0.139]    | 0.295       | 35131.0           | <b>0.001</b> |
|                   | Factor B (Critic: w/o vs w/)    | -0.004<br>[-0.056, 0.049]  | -0.012      | 32216.0           | 0.895        |
| Emotion (EM)      | Factor A (Guidance: RAF vs AVL) | -0.162<br>[-0.232, -0.091] | -0.396      | 27076.0           | <b>0.001</b> |
|                   | Factor B (Critic: w/o vs w/)    | 0.009<br>[-0.063, 0.082]   | 0.022       | 32627.5           | 0.803        |
| Suspense (SU)     | Factor A (Guidance: RAF vs AVL) | -0.044<br>[-0.081, -0.007] | -0.205      | 30895.0           | <b>0.022</b> |
|                   | Factor B (Critic: w/o vs w/)    | -0.011<br>[-0.048, 0.027]  | -0.049      | 31993.5           | 0.581        |
| Engagement (EG)   | Factor A (Guidance: RAF vs AVL) | -0.094<br>[-0.181, -0.007] | -0.189      | 29253.0           | <b>0.034</b> |
|                   | Factor B (Critic: w/o vs w/)    | 0.011<br>[-0.076, 0.098]   | 0.022       | 32687.5           | 0.804        |
| Complexity (CX)   | Factor A (Guidance: RAF vs AVL) | -0.037<br>[-0.081, 0.006]  | -0.152      | 31119.5           | 0.087        |
|                   | Factor B (Critic: w/o vs w/)    | 0.013<br>[-0.030, 0.056]   | 0.054       | 32748.5           | 0.543        |
| Overall (OV)      | Factor A (Guidance: RAF vs AVL) | -0.043<br>[-0.071, -0.014] | -0.263      | 28301.5           | <b>0.010</b> |
|                   | Factor B (Critic: w/o vs w/)    | 0.003<br>[-0.026, 0.032]   | 0.018       | 32423.0           | 0.953        |

*Note:* Mean Difference for Factor A =  $\text{Mean}_{\text{RAF}} - \text{Mean}_{\text{AVL}}$ . Mean Difference for Factor B =  $\text{Mean}_{\text{w/o\_Critic}} - \text{Mean}_{\text{w/_Critic}}$ . [95% CI] represents the 95% confidence interval of the mean difference.

Table S3: Detailed Main Effects of the  $2 \times 2$  Factorial Evaluation (**AllMean Pool, Qwen3-14B Judge**). Statistical significance is derived from the Mann-Whitney U test. Significant  $p$ -values ( $p < 0.05$ ) are highlighted in bold.

| Evaluation Metric | Analyzed Effect                 | Diff.<br>[95% CI]         | Cohen's $d$ | Statistic ( $U$ ) | $p$ -value   |
|-------------------|---------------------------------|---------------------------|-------------|-------------------|--------------|
| Relevance (RE)    | Factor A (Guidance: RAF vs AVL) | 0.000<br>[0.000, 0.000]   | 0.000       | 32294.0           | 1.000        |
|                   | Factor B (Critic: w/o vs w/)    | 0.000<br>[0.000, 0.000]   | 0.000       | 32330.0           | 1.000        |
| Coherence (CH)    | Factor A (Guidance: RAF vs AVL) | 0.000<br>[0.000, 0.000]   | 0.000       | 32294.0           | 1.000        |
|                   | Factor B (Critic: w/o vs w/)    | 0.000<br>[0.000, 0.000]   | 0.000       | 32330.0           | 1.000        |
| Emotion (EM)      | Factor A (Guidance: RAF vs AVL) | 0.000<br>[0.000, 0.000]   | 0.000       | 32294.0           | 1.000        |
|                   | Factor B (Critic: w/o vs w/)    | 0.000<br>[0.000, 0.000]   | 0.000       | 32330.0           | 1.000        |
| Suspense (SU)     | Factor A (Guidance: RAF vs AVL) | 0.020<br>[-0.047, 0.086]  | 0.052       | 32833.5           | 0.597        |
|                   | Factor B (Critic: w/o vs w/)    | 0.006<br>[-0.060, 0.072]  | 0.016       | 32505.0           | 0.864        |
| Engagement (EG)   | Factor A (Guidance: RAF vs AVL) | 0.030<br>[0.009, 0.050]   | 0.241       | 33258.0           | <b>0.007</b> |
|                   | Factor B (Critic: w/o vs w/)    | -0.007<br>[-0.028, 0.015] | -0.053      | 32117.5           | 0.553        |
| Complexity (CX)   | Factor A (Guidance: RAF vs AVL) | 0.082<br>[0.022, 0.142]   | 0.237       | 34848.5           | <b>0.007</b> |
|                   | Factor B (Critic: w/o vs w/)    | -0.019<br>[-0.079, 0.042] | -0.053      | 31701.0           | 0.507        |
| Overall (OV)      | Factor A (Guidance: RAF vs AVL) | 0.022<br>[0.006, 0.038]   | 0.235       | 34584.0           | 0.055        |
|                   | Factor B (Critic: w/o vs w/)    | -0.003<br>[-0.020, 0.013] | -0.034      | 32989.5           | 0.582        |

*Note:* Mean Difference for Factor A =  $\text{Mean}_{\text{RAF}} - \text{Mean}_{\text{AVL}}$ . Mean Difference for Factor B =  $\text{Mean}_{\text{w/o\_Critic}} - \text{Mean}_{\text{w\_Critic}}$ . [95% CI] represents the 95% confidence interval of the mean difference.

Table S4: Detailed Main Effects of the  $2 \times 2$  Factorial Evaluation (**Top-3 Pool, Mistral-7B** Judge). Statistical significance is derived from the Mann-Whitney U test. Significant  $p$ -values ( $p < 0.05$ ) are highlighted in bold.

| Evaluation Metric | Analyzed Effect                 | Diff.<br>[95% CI]          | Cohen's $d$ | Statistic ( $U$ ) | $p$ -value   |
|-------------------|---------------------------------|----------------------------|-------------|-------------------|--------------|
| Relevance (RE)    | Factor A (Guidance: RAF vs AVL) | -0.167<br>[-0.269, -0.065] | -0.438      | 4846              | <b>0.008</b> |
|                   | Factor B (Critic: w/o vs w/)    | 0.009<br>[-0.095, 0.114]   | 0.024       | 5753              | 0.833        |
| Coherence (CH)    | Factor A (Guidance: RAF vs AVL) | 0.185<br>[-0.084, 0.454]   | 0.185       | 6419              | <b>0.013</b> |
|                   | Factor B (Critic: w/o vs w/)    | 0.019<br>[-0.252, 0.289]   | 0.018       | 5581.5            | 0.290        |
| Emotion (EM)      | Factor A (Guidance: RAF vs AVL) | -0.111<br>[-0.186, -0.036] | -0.398      | 4940              | <b>0.005</b> |
|                   | Factor B (Critic: w/o vs w/)    | 0.083<br>[0.008, 0.159]    | 0.296       | 6581              | <b>0.018</b> |
| Suspense (SU)     | Factor A (Guidance: RAF vs AVL) | -0.147<br>[-0.264, -0.029] | -0.335      | 4959              | <b>0.024</b> |
|                   | Factor B (Critic: w/o vs w/)    | 0.073<br>[-0.046, 0.191]   | 0.164       | 6198              | 0.346        |
| Engagement (EG)   | Factor A (Guidance: RAF vs AVL) | -0.148<br>[-0.275, -0.022] | -0.314      | 4872              | <b>0.020</b> |
|                   | Factor B (Critic: w/o vs w/)    | 0.000<br>[-0.128, 0.128]   | 0.000       | 5848              | 0.970        |
| Complexity (CX)   | Factor A (Guidance: RAF vs AVL) | -0.037<br>[-0.098, 0.024]  | -0.163      | 5518.0            | 0.211        |
|                   | Factor B (Critic: w/o vs w/)    | 0.037<br>[-0.024, 0.098]   | 0.163       | 6146.0            | 0.211        |
| Overall (OV)      | Factor A (Guidance: RAF vs AVL) | -0.092<br>[-0.144, -0.040] | -0.477      | 4471              | <b>0.002</b> |
|                   | Factor B (Critic: w/o vs w/)    | 0.030<br>[-0.023, 0.083]   | 0.153       | 6148.5            | 0.478        |

*Note:* Mean Difference for Factor A =  $\text{Mean}_{\text{RAF}} - \text{Mean}_{\text{AVL}}$ . Mean Difference for Factor B =  $\text{Mean}_{\text{w/o\_Critic}} - \text{Mean}_{\text{w/\_Critic}}$ . [95% CI] represents the 95% confidence interval of the mean difference.

Table S5: Detailed Main Effects of the  $2 \times 2$  Factorial Evaluation (**Top-3 Pool**, **Gemma3-12B** Judge). Statistical significance is derived from the Mann-Whitney U test. Significant  $p$ -values ( $p < 0.05$ ) are highlighted in bold.

| Evaluation Metric | Analyzed Effect                 | Diff.<br>[95% CI]          | Cohen's $d$ | Statistic ( $U$ ) | $p$ -value     |
|-------------------|---------------------------------|----------------------------|-------------|-------------------|----------------|
| Relevance (RE)    | Factor A (Guidance: RAF vs AVL) | 0.000<br>[-0.026, 0.026]   | 0.000       | 5832.0            | 1.000          |
|                   | Factor B (Critic: w/o vs w/)    | 0.000<br>[-0.026, 0.026]   | 0.000       | 5832.0            | 1.000          |
| Coherence (CH)    | Factor A (Guidance: RAF vs AVL) | 0.074<br>[-0.004, 0.152]   | 0.257       | 6264.0            | 0.061          |
|                   | Factor B (Critic: w/o vs w/)    | 0.019<br>[-0.060, 0.097]   | 0.064       | 5940.0            | 0.641          |
| Emotion (EM)      | Factor A (Guidance: RAF vs AVL) | -0.241<br>[-0.368, -0.113] | -0.507      | 4428.0            | < <b>0.001</b> |
|                   | Factor B (Critic: w/o vs w/)    | 0.019<br>[-0.113, 0.150]   | 0.038       | 5940.0            | 0.782          |
| Suspense (SU)     | Factor A (Guidance: RAF vs AVL) | -0.093<br>[-0.170, -0.015] | -0.322      | 5292.0            | <b>0.019</b>   |
|                   | Factor B (Critic: w/o vs w/)    | -0.019<br>[-0.097, 0.060]  | -0.064      | 5724.0            | 0.641          |
| Engagement (EG)   | Factor A (Guidance: RAF vs AVL) | -0.130<br>[-0.244, -0.016] | -0.305      | 5076.0            | <b>0.026</b>   |
|                   | Factor B (Critic: w/o vs w/)    | 0.019<br>[-0.097, 0.134]   | 0.043       | 5940.0            | 0.752          |
| Complexity (CX)   | Factor A (Guidance: RAF vs AVL) | -0.019<br>[-0.069, 0.032]  | -0.098      | 5724.0            | 0.474          |
|                   | Factor B (Critic: w/o vs w/)    | 0.000<br>[-0.051, 0.051]   | 0.000       | 5832.0            | 1.000          |
| Overall (OV)      | Factor A (Guidance: RAF vs AVL) | -0.068<br>[-0.105, -0.031] | -0.489      | 4469.5            | <b>0.001</b>   |
|                   | Factor B (Critic: w/o vs w/)    | 0.006<br>[-0.032, 0.045]   | 0.043       | 5991.0            | 0.703          |

*Note:* Mean Difference for Factor A =  $\text{Mean}_{\text{RAF}} - \text{Mean}_{\text{AVL}}$ . Mean Difference for Factor B =  $\text{Mean}_{\text{w/o\_Critic}} - \text{Mean}_{\text{w/\_Critic}}$ . [95% CI] represents the 95% confidence interval of the mean difference.

Table S6: Detailed Main Effects of the  $2 \times 2$  Factorial Evaluation (**Top-3 Pool, Qwen3-14B** Judge). Statistical significance is derived from the Mann-Whitney U test. Significant  $p$ -values ( $p < 0.05$ ) are highlighted in bold.

| Evaluation Metric | Analyzed Effect                 | Diff.<br>[95% CI]         | Cohen's $d$ | Statistic ( $U$ ) | $p$ -value |
|-------------------|---------------------------------|---------------------------|-------------|-------------------|------------|
| Relevance (RE)    | Factor A (Guidance: RAF vs AVL) | 0.000<br>[0.000, 0.000]   | 0.000       | 5832.0            | 1.000      |
|                   | Factor B (Critic: w/o vs w/)    | 0.000<br>[0.000, 0.000]   | 0.000       | 5832.0            | 1.000      |
| Coherence (CH)    | Factor A (Guidance: RAF vs AVL) | 0.000<br>[0.000, 0.000]   | 0.000       | 5832.0            | 1.000      |
|                   | Factor B (Critic: w/o vs w/)    | 0.000<br>[0.000, 0.000]   | 0.000       | 5832.0            | 1.000      |
| Emotion (EM)      | Factor A (Guidance: RAF vs AVL) | 0.000<br>[0.000, 0.000]   | 0.000       | 5832.0            | 1.000      |
|                   | Factor B (Critic: w/o vs w/)    | 0.000<br>[0.000, 0.000]   | 0.000       | 5832.0            | 1.000      |
| Suspense (SU)     | Factor A (Guidance: RAF vs AVL) | -0.015<br>[-0.112, 0.081] | -0.043      | 5732.0            | 0.725      |
|                   | Factor B (Critic: w/o vs w/)    | 0.053<br>[-0.044, 0.149]  | 0.146       | 6147.0            | 0.266      |
| Engagement (EG)   | Factor A (Guidance: RAF vs AVL) | 0.009<br>[-0.009, 0.028]  | 0.136       | 5886.0            | 0.322      |
|                   | Factor B (Critic: w/o vs w/)    | -0.009<br>[-0.028, 0.009] | -0.136      | 5778.0            | 0.322      |
| Complexity (CX)   | Factor A (Guidance: RAF vs AVL) | 0.049<br>[-0.056, 0.155]  | 0.125       | 6154.0            | 0.290      |
|                   | Factor B (Critic: w/o vs w/)    | -0.031<br>[-0.137, 0.075] | -0.078      | 5642.5            | 0.534      |
| Overall (OV)      | Factor A (Guidance: RAF vs AVL) | 0.007<br>[-0.015, 0.029]  | 0.087       | 5922.0            | 0.798      |
|                   | Factor B (Critic: w/o vs w/)    | 0.002<br>[-0.020, 0.024]  | 0.025       | 6058.5            | 0.519      |

*Note:* Mean Difference for Factor A =  $\text{Mean}_{\text{RAF}} - \text{Mean}_{\text{AVL}}$ . Mean Difference for Factor B =  $\text{Mean}_{\text{w/o\_Critic}} - \text{Mean}_{\text{w\_Critic}}$ . [95% CI] represents the 95% confidence interval of the mean difference.

## S1.2 Detailed Affective Salience and Intensity Statistics

Table S7 presents the statistical testing on Affective Salience (AS) and the Proportion of High-Salience (HSP). To ensure statistical rigor, the choice between Welch’s t-test and the Mann-Whitney U test was determined dynamically based on the Shapiro-Wilk test for normality.

Table S7: Detailed Statistical Tests on Affective Salience and Intensity. Significance levels: ns ( $p \geq 0.05$ ), \* ( $p < 0.05$ ), \*\* ( $p < 0.01$ ), \*\*\* ( $p < 0.001$ ).

| Metric                                     | Comparison                       | Mean A | Mean B | Diff.<br>[95% CI]          | Cohen’s $d$ | Test Used      | $p$ -value           |
|--------------------------------------------|----------------------------------|--------|--------|----------------------------|-------------|----------------|----------------------|
| Valence<br>Affective<br>Salience (AS)      | RAF_w/o_Critic vs RAF_w/_Critic  | 0.100  | 0.096  | 0.004<br>[0.000, 0.008]    | 0.269       | Mann-Whitney U | <b>0.040</b> (*)     |
|                                            | AVL_w/o_Critic vs AVL_w/_Critic  | 0.107  | 0.104  | 0.003<br>[-0.001, 0.007]   | 0.211       | Welch’s t      | 0.085 (ns)           |
|                                            | RAF_w/o_Critic vs AVL_w/o_Critic | 0.100  | 0.107  | -0.007<br>[-0.011, -0.003] | -0.428      | Mann-Whitney U | <b>0.003</b> (**)    |
|                                            | RAF_w/_Critic vs AVL_w/_Critic   | 0.096  | 0.104  | -0.008<br>[-0.011, -0.004] | -0.497      | Welch’s t      | < <b>0.001</b> (***) |
| Arousal<br>Affective<br>Salience (AS)      | RAF_w/o_Critic vs RAF_w/_Critic  | 0.235  | 0.236  | -0.002<br>[-0.004, 0.001]  | -0.148      | Welch’s t      | 0.258 (ns)           |
|                                            | AVL_w/o_Critic vs AVL_w/_Critic  | 0.234  | 0.235  | -0.000<br>[-0.003, 0.002]  | -0.041      | Mann-Whitney U | 0.895 (ns)           |
|                                            | RAF_w/o_Critic vs AVL_w/o_Critic | 0.235  | 0.234  | 0.001<br>[-0.002, 0.003]   | 0.071       | Mann-Whitney U | 0.752 (ns)           |
|                                            | RAF_w/_Critic vs AVL_w/_Critic   | 0.236  | 0.235  | 0.002<br>[-0.001, 0.005]   | 0.167       | Welch’s t      | 0.197 (ns)           |
| Valence<br>High-Salience<br>Proportion (%) | RAF_w/o_Critic vs RAF_w/_Critic  | 23.32  | 21.85  | 1.47<br>[0.07, 2.87]       | 0.268       | Mann-Whitney U | 0.057 (ns)           |
|                                            | AVL_w/o_Critic vs AVL_w/_Critic  | 25.81  | 24.41  | 1.39<br>[-0.18, 2.97]      | 0.213       | Mann-Whitney U | 0.093 (ns)           |
|                                            | RAF_w/o_Critic vs AVL_w/o_Critic | 23.32  | 25.81  | -2.48<br>[-3.95, -1.02]    | -0.408      | Mann-Whitney U | <b>0.004</b> (**)    |
|                                            | RAF_w/_Critic vs AVL_w/_Critic   | 21.85  | 24.41  | -2.56<br>[-4.07, -1.05]    | -0.423      | Welch’s t      | < <b>0.001</b> (***) |
| Arousal<br>High-Salience<br>Proportion (%) | RAF_w/o_Critic vs RAF_w/_Critic  | 82.65  | 82.80  | -0.16<br>[-1.11, 0.80]     | -0.042      | Welch’s t      | 0.751 (ns)           |
|                                            | AVL_w/o_Critic vs AVL_w/_Critic  | 82.16  | 82.55  | -0.39<br>[-1.29, 0.52]     | -0.103      | Mann-Whitney U | 0.417 (ns)           |
|                                            | RAF_w/o_Critic vs AVL_w/o_Critic | 82.65  | 82.16  | 0.49<br>[-0.35, 1.33]      | 0.141       | Mann-Whitney U | 0.432 (ns)           |
|                                            | RAF_w/_Critic vs AVL_w/_Critic   | 82.80  | 82.55  | 0.26<br>[-0.76, 1.27]      | 0.064       | Mann-Whitney U | 0.754 (ns)           |

Note: The statistical test (Welch’s t or Mann-Whitney U) was selected based on the outcome of the Shapiro-Wilk test for normality on the respective distributions. [95% CI] represents the 95% confidence interval of the mean difference.

## Section S2: Prompt Engineering

This section shows the condensed key instructions in the outline and scripts generation prompts. The complete system and user prompts used in the **CRAVE** framework are highly structured. To ensure the readability of this document while maintaining full experimental transparency, we provide:

1. **Core Logic Templates:** The fundamental instructional blocks for the two main mechanisms.
2. **Full Repository:** The raw prompt files, source code, and additional qualitative samples are available at the supplementary repository.

### S2.1 AVL Prompts

Figure S1 shows a condensed representation of the core AVL prompts in the outline and scripts generation phases.

### S2.2 Critic Prompts

Figure S2 shows a condensed representation of the core Critic evaluation prompts in the outline and scripts generation phases.



### Full Dual-Phase AVL Execution Prompt

#### Core Instruction:

You will be provided with [Valence] and [Arousal] coordinates. EXECUTE this state strictly.

#### 1. Valence Mapping (Plot Direction & Emotional Polarity):

- *Level 1 (Very Negative)* → **Outline:** Major setbacks, tragedies. **Script:** Lexical despair, terror.
- *Level 2 (Negative)* → **Outline:** Minor conflicts, obstacles. **Script:** Tension, sadness, anxiety.
- *Level 3 (Neutral)* → **Outline:** Transitions, balanced events. **Script:** Calm, neutral tone.
- *Level 4 (Positive)* → **Outline:** Small victories, progress. **Script:** Relief, hope, optimism.
- *Level 5 (Very Positive)* → **Outline:** Major resolutions, catharsis. **Script:** Lexical triumph.

#### 2. Arousal Mapping (Event Density & Emotional Intensity):

- *Level 1 (Very Low)* → **Outline:** Reflective moments. **Script:** Linguistic silence, stagnation.
- *Level 2 (Low)* → **Outline:** Slow buildup. **Script:** Introspective, measured pacing.
- *Level 3 (Mid)* → **Outline:** Steady progression. **Script:** Normal conversational pacing.
- *Level 4 (High)* → **Outline:** Fast developments. **Script:** Urgency, increased pressure.
- *Level 5 (Very High)* → **Outline:** Climactic action, twists. **Script:** Chaotic pacing, panic.

#### Execution Target:

*Outlining:* Embed the tone into the causal narrative structure.

*Scripting:* Adopt the state through specific word choice and pacing.

Figure S1: The explicit  $5 \times 5$  affective constraints provided to the LLM. The mapping dictates macro-level plot events during outlining and micro-level stylistic choices during script generation. The complete prompts for both outlining and scripting stages are available in the Supplementary Material repository.

### Simplified Critic Evaluation Prompt

**Role:** You are an expert evaluation critic judging visual novel generation drafts. Do NOT rewrite; strictly evaluate based on the following dimensions:

#### 1. Intrinsic Narrative Quality:

Ensure the text is engaging, logical, and character-driven within its [GENRE] and [THEME]. It must not be an empty vehicle for emotions.

#### 2. Affective Execution & Progression:

Treat [Valence] (emotional tone) and [Arousal] (pacing/energy) as strict narrative signals.

- *Valence check:* Lexical sentiment and plot setbacks must match the target.
- *Arousal check:* Linguistic urgency, action frequency, and structural density must match the target.

#### 3. Narrative Justification:

The required emotional and pacing shifts must feel meticulously controlled and fully justified by actual narrative events.

#### [PENALIZE]:

- Emotional shifts that feel forced or unearned.
- Contradictions between tone and pacing (e.g., calm tone with chaotic arousal).

Figure S2: A condensed representation of the core evaluation instructions provided to the Critic Agent. The complete evaluation prompts for both outlining and scripting stages are available in the Supplementary Material repository.

### Evaluation Prompt

Theme: {{THEME}}; Genre: {{GENRE}}

Target Story: {{STORY}}

Rate the visual novel story on a scale from 1 to 5 on the following criteria and explain your answer:

1. **Relevance** (how well the story matches its Theme and Genre)
2. **Coherence** (how much the story makes sense)
3. **Empathy** (how well the reader understood the character's emotions)
4. **Surprise** (how surprising the end of story was)
5. **Engagement** (how much the reader engaged with the story)
6. **Complexity** (how elaborate the story is)

Output the ratings and explanations strictly in JSON format.

Structure your response as follows (use these exact keys):

```
{
  "RE_Score": 1-5,
  "RE_Explanation": "Why...",
  "CH_Score": 1-5,
  "CH_Explanation": "Why...",
  "EM_Score": 1-5,
  "EM_Explanation": "Why...",
  "SU_Score": 1-5,
  "SU_Explanation": "Why...",
  "EG_Score": 1-5,
  "EG_Explanation": "Why...",
  "CX_Score": 1-5,
  "CX_Explanation": "Why..."
}
```

Figure S3: Evaluation prompt used in LLM-as-a-Judge Evaluation

## Section S3: LLM-as-a-Judge Evaluation Criteria

As introduced in **Section IV-B** of the main manuscript, we employ an LLM-as-a-Judge methodology to quantitatively assess the qualitative aspects of the generated visual novel branches. Figure S3 details the exact prompt utilized to evaluate the narrative across the six distinct dimensions presented in our results. The model is forced to output both a numerical score and a reasoning explanation to ensure the interpretability of the evaluation data.